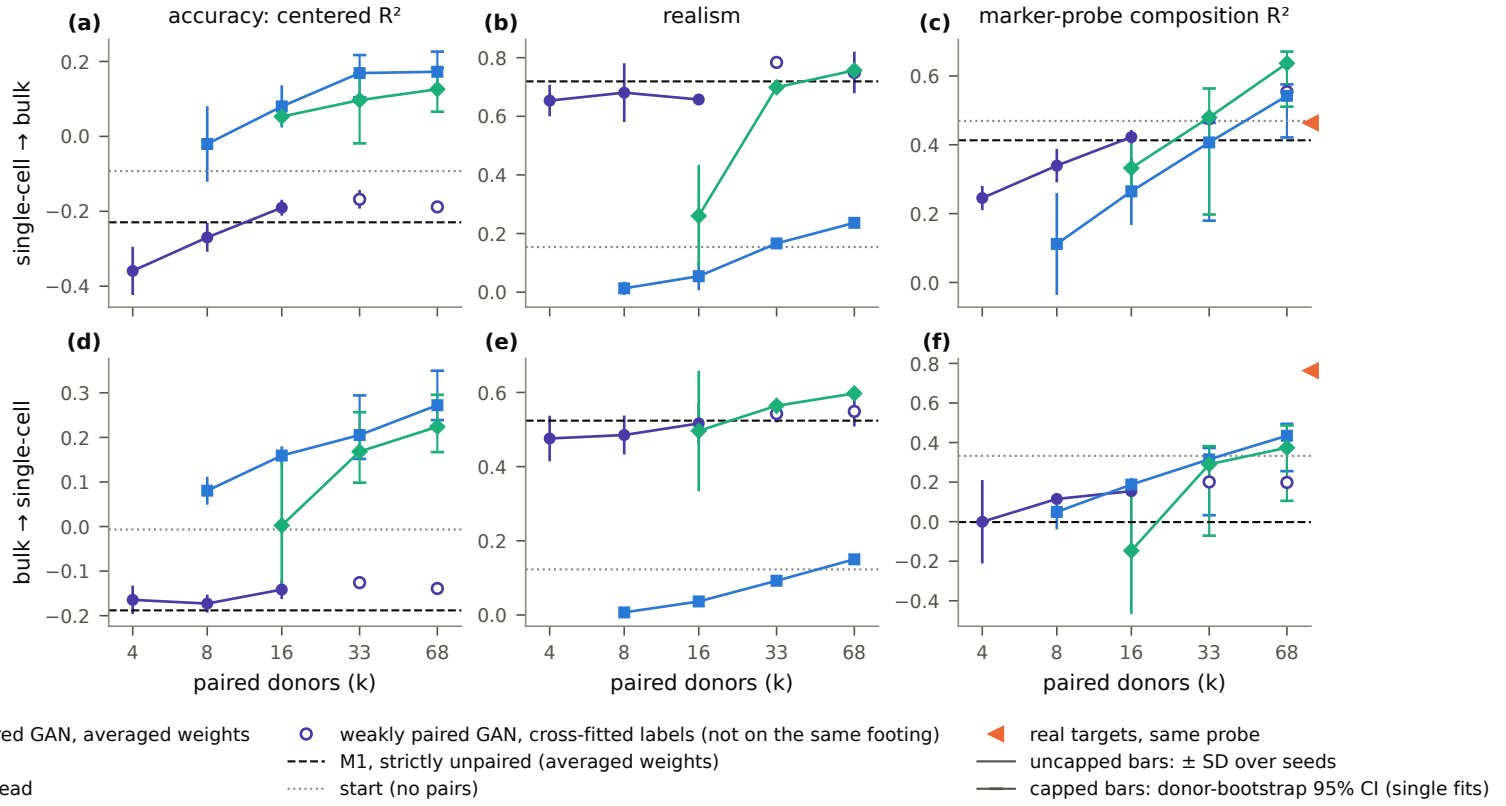
(a)
accuracy: centered R²
(b)
realism
(c)
marker-probe composition R²
single-cell → bulk
(d)
(e)
(f)
bulk → single-cell
paired donors (k)
paired donors (k)
paired donors (k)
weakly paired GAN, averaged weights
ridge
ridge + spread
weakly paired GAN, cross-fitted labels (not on the same footing)
M1, strictly unpaired (averaged weights)
start (no pairs)
real targets, same probe
uncapped bars: ± SD over seeds
capped bars: donor-bootstrap 95% CI (single fits)